Faculty of Engineering and the Built Environment

Department of Electrical Engineering

2026 Course Handout: **EEE3096S**

Course Name:	**EEE3096S: EMBEDDED SYSTEMS II**
SAQA Credits:	16
Pre-requisites:	EEE2046F (Embedded Systems I)
Co-requisites:	None

Course convenor:	A/Prof Simon Winberg
Email address:	simon.winberg@uct.ac.za
Office location:	Room 6.13, Menzies Building
Consultation hours:	Tue/Wed after lecture or email to request a consultation
Course lecturer:	A/Prof Simon Winberg & Mr Travimadox Webb
Teaching assistant:	Maisha Magavha

Lecture venue:	JORDAN LT3A
Lecture days and time:	**Lecture days:** Tuesdays, Wednesdays at 10h00 – 11h45 Four lecture slots per week, but one of these will likely be used as a tutorial/practical/study/consultation slot every week, based on student needs and course content.
Laboratory venue	**White Lab, Menzies Building**
Laboratory days and times	To be determined … Tutors will be available in laboratory venues to help complete the practicals. Completion of practicals must be demonstrated to tutors on campus during these sessions. An announcement will be made to indicate when a new practical is released.

Course objectives

This course integrates aspects of embedded systems and computer architecture by building on EEE2046S (2nd year Embedded Systems I), which provided introductory knowledge of digital electronics, microcontrollers, and C programming. Embedded Systems II introduces students to microprocessor architecture fundamentals, memory technologies, embedded communication protocols, Assembly programming, state machines, and ADCs and DACs. The course additionally expands students' knowledge of C programming for embedded systems, peripherals and digital components used in cyber-physical interfaces (such as timers, GPIO, RAM and EEPROM), and theories relating to signal processing and hardware-software interfaces.

Practicals will involve using a single-board computer embedded platform (the "UCT STM development board"), and include work in hardware-software interfacing, C and Assembly programming, and systems design. The course is broken into various modules and will include practical exercises to reinforce students' understanding of core concepts.

There is no prescribed textbook for the course, and a full set of course notes will be made available to students. For students who wish to pursue further study on the material, a recommended textbook for this course is: "Computer Organization and Design ARM Edition: The Hardware Software Interface" (The Morgan Kaufmann Series in Computer Architecture and Design) 1st by Ed. D. Patterson and J. Hennessy (available on Amazon at https://www.amazon.com/Computer-Organization-Design-ARM-Architecture/dp/0128017333).

Learning outcomes

Students successfully completing this course will have the following:	Exit level	LO1	LO2	LO3	LO4	LO5	LO6	LO7	LO8	LO9	LO10	LO11
A. Knowledge (Information plus understanding)												
1. Knowledge of the design lifecycle of an embedded system	N			X		X						
2. Knowledge of computer and microprocessor architectures pertinent to use in low-power, wireless, and or otherwise edge devices	N			X		X						
3. Knowledge of techniques for the analysis of embedded software designs and performance	N			X		X						
4. Knowledge of communication and interfacing techniques for embedded systems	N					X						
5. Knowledge of current changes and trends regarding how embedded systems are being used in a wide range of sectors of society	N					X						
6. Knowledge of how ADCs and DACs function and their use in embedded systems	N					X						
B. Skills (Application of knowledge)												
1. Modelling embedded system operation	N		X	X		X						
2. State machine design	N		X	X		X						
3. Techniques to model and analyse the performance of computer architecture and embedded systems	N		X	X		X						
4. Use of a cross-compiler	N		X			X						
5. Practical use of standard embedded systems communication protocol	N		X			X						
6. Debugging techniques for embedded systems	N		X			X						
7. Understanding and writing basic Assembly language programs	N		X			X						
8. Developing Assembly-based solutions for embedded systems	N		X			X						

C. Values and attitudes												
1. Recognition of characteristics of quality embedded system designs	N						X					
2. Importance of design documentation	N			X			X					
3. Value of thorough testing	N			X								
4. Software engineering design principles for real-time systems	N			X								
5. Appreciation of the value of teamwork in embedded system design & implementation	N						X		X			
6. Appreciation for the use of appropriate tools for collaborative and complex design work	N						X		X			

Detailed course content

As described in the objectives, the course extends students' knowledge in design and use of embedded systems by introducing new theory, expanding their knowledge in a range of areas, and providing a learning environment within which to practically apply both theory and knowledge. The course is delivered in multiple modules, each focused on a core theme.

Details of the six learning modules of this course is provided below:

1. **Embedded Systems**
    a. Embedded Systems (ES) and Cyber-physical systems (CPS)
    b. Real-time Systems (RTS)
    c. Development, Execution and Runtime Environments
    d. Embedded C programming
    e. Make and compiler flags (cmake optional reading)
    f. Debugging
    g. Optimisation and Benchmarking
    h. State Machines
2. **Assembly and C Programming**
    a. Intro to ARM
    b. RISC vs. CISC processors
    c. ARM, Thumb, and Thumb-2 instruction sets
    d. Exceptions and Branching
    e. The Stack
    f. Loading, Storing, Branching
    g. Subroutines
    h. Peripherals
    i. Exceptions and interrupts
    j. Timers
3. **Embedded Communications**
    a. Synchronous and Asynchronous
    b. Serial and Parallel
    c. SPI
    d. I2C
    e. UART, RS-232
    f. PWM, debouncing
4. **Sampling, ADCs, Signal Generation, and DACs**
    a. Signal processing
    b. Information theory and DAC signal generation
    c. ADCs and Sampling
    d. Filtering
5. **Computer Architecture** (time permitting)
    a. CPU Architecture
    b. ALU and Datapath
    c. Buses and Memory

Knowledge areas

Maths Sciences	Natural Sciences	Engineering Sciences	Design & Synthesis	Complementary Studies
-	-	60	40	-

Learning environment

Multimodal. Combination of laboratory work involving hardware, lecture sessions, and course material (course notes, other learning resources, etc). Students will be expected to engage with material in various forms in a variety of different ways, but have much flexibility in where and how this material is accessed.

Suggested time allocation

Learning Activity	**No./ week**	**Time in hours**	**Contact time Multiplier**	**Total no of hours**
Number of lectures per week (for 12 weeks)	2	1.5	2	72
Number of tutorials per week (for 12 weeks)	0	0	0	0
Number of practicals (for 12 weeks)	5	2	3	30
Total other contact periods				
Total assignment non-contact hours				45
Assessment hours (Tests, Exam)		5	3	15
Number of weeks the course lasts (extra week for revision, extensions, public holidays)	13			
Total hours				**162**

General assessment strategy

Assessment Task	%	The following DP rules apply:
Labs	35%	• Completion of all practical assignments. • To receive DP for this course, the student must achieve a coursework average (i.e. all assignments besides exam) of 40% or above. **Late penalties will be applied for assignments (see assignment for further detail on specific penalties; default is 10%/day).**
Class Test	10%	
Hackathon	5%	
Exam	50%	
Total	**100%**	

Books/Reading Materials/Notes

- Optional textbook: “Computer Organization and Design ARM Edition: The Hardware Software Interface” (The Morgan Kaufmann Series in Computer Architecture and Design) 1st Edition. Authors: David A. Patterson and John L. Hennessy.
  On Amazon at: https://www.amazon.com/Computer-Organization-Design-ARM-Architecture/dp/0128017333
- Supplementary (optional) textbook: “Embedded System Design: Embedded Systems Foundations of Cyber-Physical Systems, and the Internet of Things” 3rd Edition by Peter Marwedel. Publisher: Springer. 2018.
- Official course notes available on Amathuba
- Amathuba: See the EEE3095/6S site

Absence: If a class test is missed, the final exam will be scaled up in its weighting to accommodate this change. E.g., missing a class test that counts for 10% of the final mark will result in the student's exam being weighted 10% extra in the final mark.

Academic dishonesty: Plagiarism is a very serious offence and usually leads to disciplinary action that could include expulsion from the university. Therefore, recognise the work of others in any submission. Details of referencing methods are widely available on the Web. A non-plagiarism declaration must be submitted with all work submitted for marking.